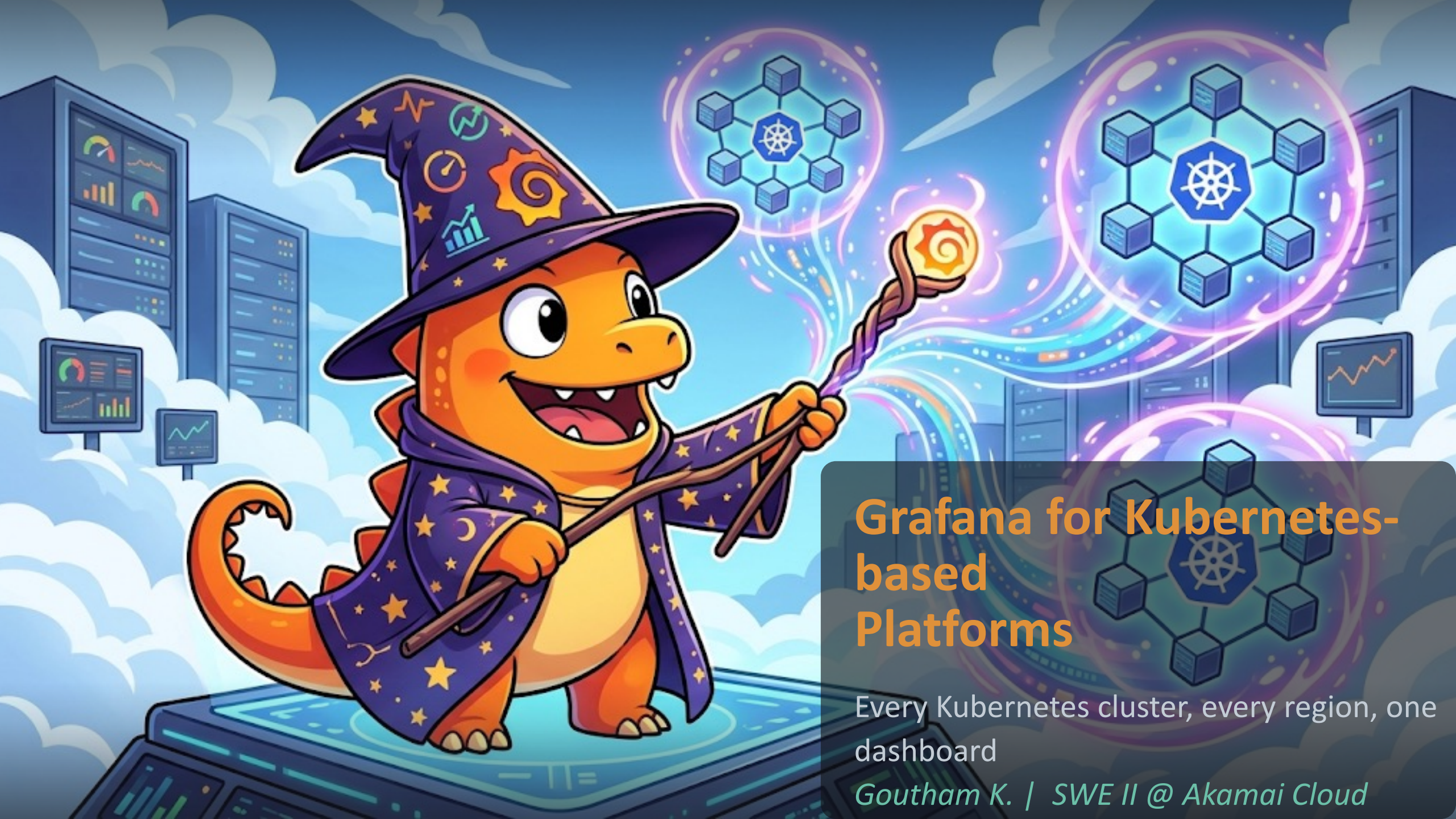




Grafana for Kubernetes-based Platforms

Every Kubernetes cluster, every region, one dashboard

Goutham K. | SWE II @ Akamai Cloud

What we'll talk about.

01 A cluster is stuck. Which one, where, and whose?

02 What an operator actually does

03 Management clusters, tenant clusters

04 Turning conditions into time series

05 Demo time!

Q Questions

THE FLEET

1,000s

tenant K8s clusters

50

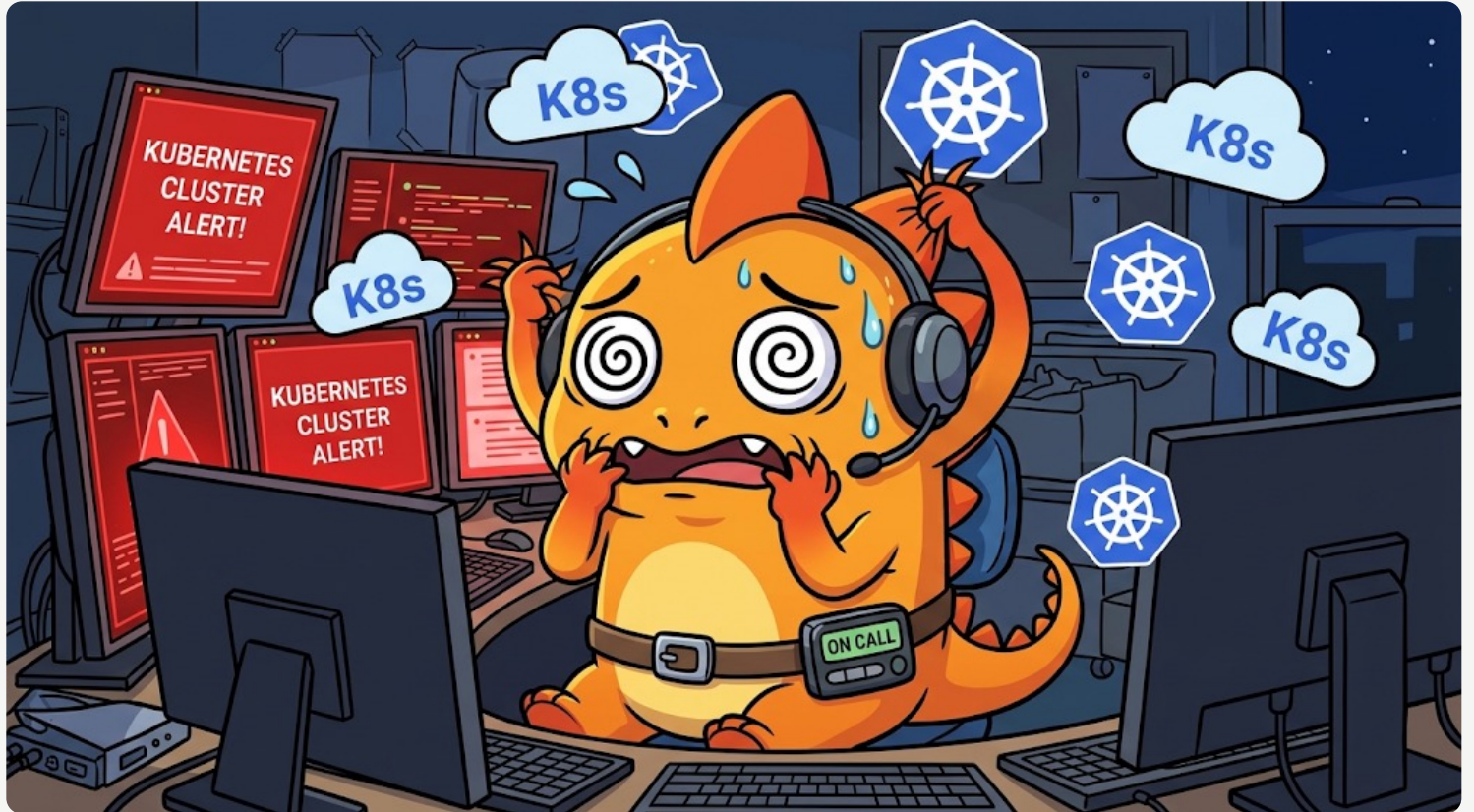
regions

50

engineers/SREs on the platform team

The page comes in at 02:40

"Hey Grot, Cluster
tuvbkdh1l4he8s5lg is
cooked.Whatup?"

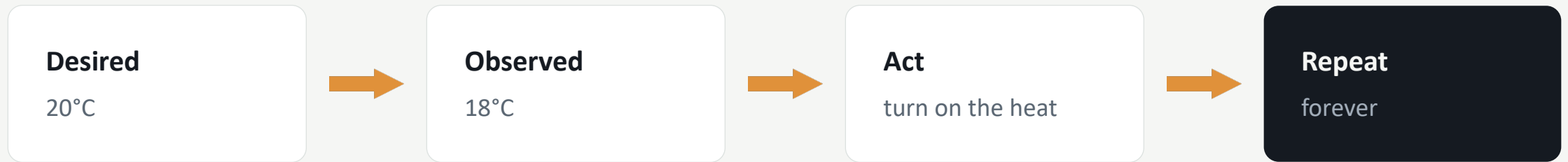


What is a Kubernetes operator?

A program that watches something you declared, compares it with what actually exists, and closes the gap — over and over, for as long as it runs.

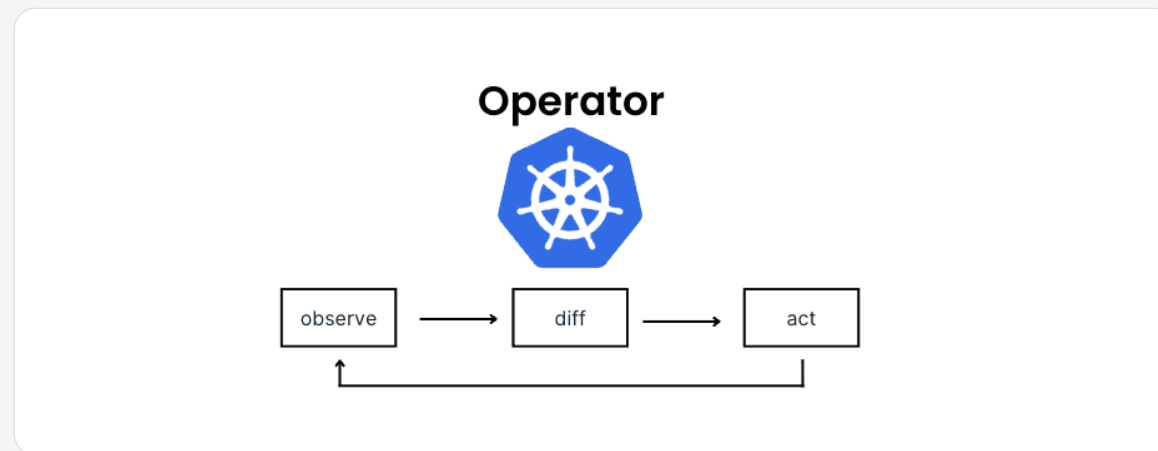
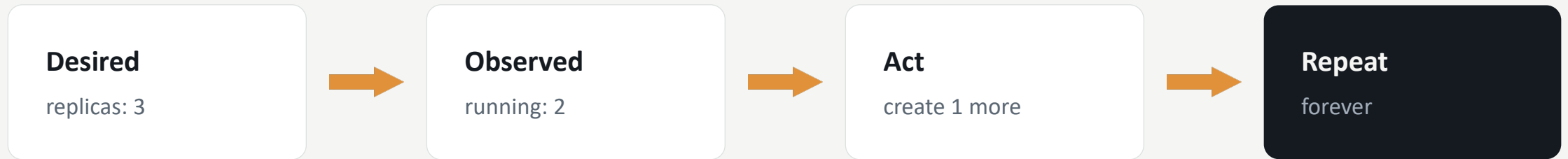
You already own one of these. It's mounted on a wall in your house.

A thermostat is a controller

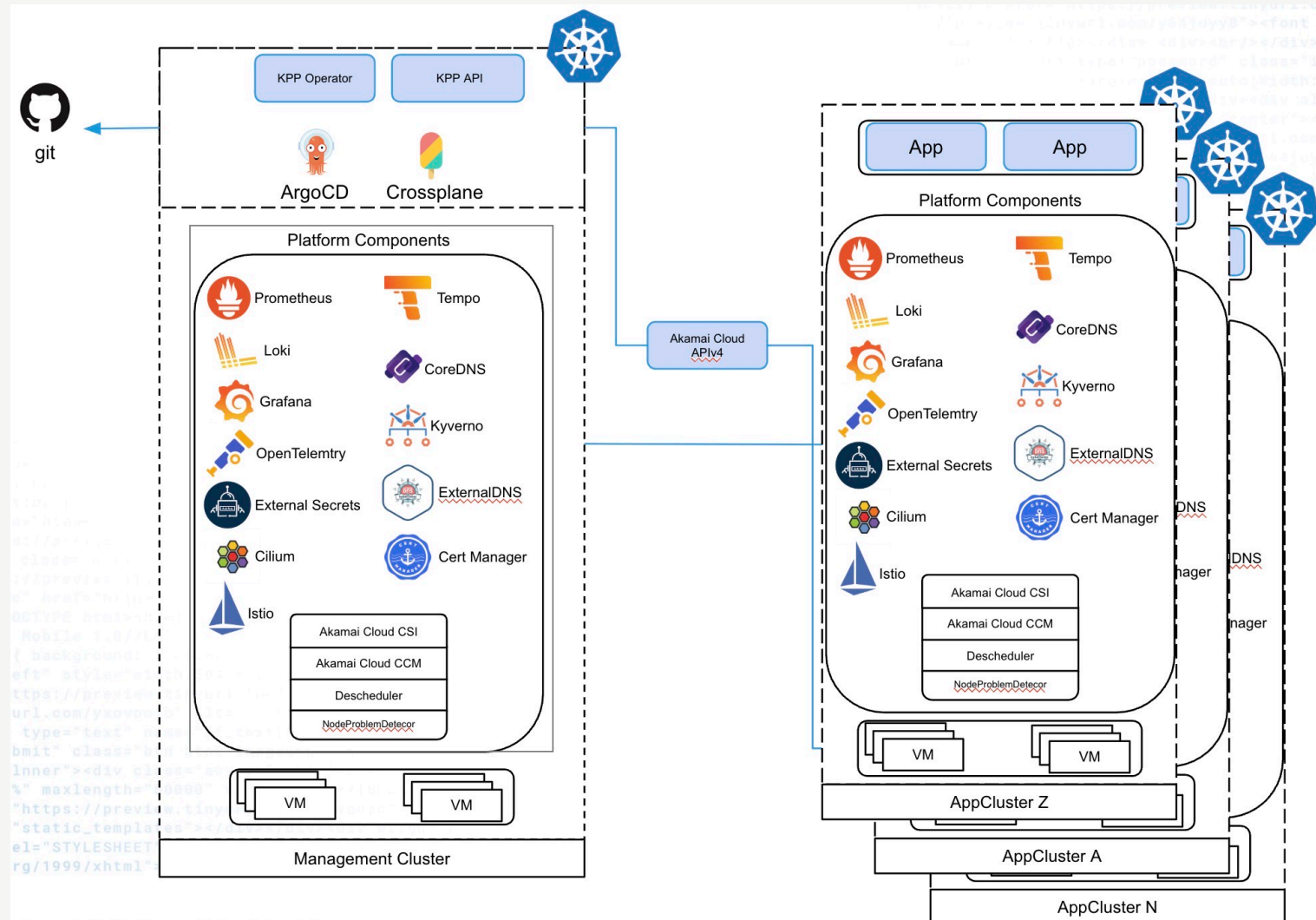


You never tell it to turn on the heat. You tell it what you want, and it closes the gap.

Same loop, with Kubernetes words on it



One management cluster, many tenant clusters



~~Is it ready?~~

Where is it stuck?

Reconciling a cluster has stages

A Deployment reconciles in milliseconds. This takes minutes to hours.

InfrastructureReady

APITokenReady

ArgoAppReady

Synced

Each one is a flag the operator writes back onto the resource, every loop.

This is where the operator writes what it knows

```
status:  
  conditions:  
    - type: InfrastructureReady  
      status: "True"  
    - type: ArgoAppReady  
      status: "False"  
      reason: SyncFailed
```

Your operator already knows why it's stuck. It just never leaves the cluster.

PATTERN 1

Identity goes in labels. The value is meaningless.

```
tenant_cluster_info{cluster_uid, name, projectId, region, version} = 1
```

Cardinality is inventory

```
count(tenant_cluster_info)
```

And it's a join table

Anything keyed on cluster_uid picks up region, version, owner

same as metrics like: kube_pod_info, node_uname_info, argocd_app_info.

Condition is a *label*, not a metric name

Do this

```
tenant_cluster_status{  
  cluster_uid,  
  condition  
} = 0 | 1
```

Not this

```
tenant_cluster_infra_ready  
tenant_cluster_argo_ready  
tenant_cluster_token_ready  
tenant_cluster_synced
```

Add a fifth condition to the operator. Option A costs you **nothing** — it's a new column, automatically. Option B costs a metric, a panel, a query and an alert rule.

ALL OF IT, IN GO

The whole exporter is one GaugeVec

Declare it once

```
var clusterStatus = prometheus.NewGaugeVec(
    prometheus.GaugeOpts{
        Name: "tenant_cluster_status",
        Help: "Condition, 1 = True",
    },
    []string{
        "cluster_uid", "name", "condition",
    },
)

func init() {
    metrics.Registry.MustRegister(clusterStatus)
}
```

Set it every reconcile

```
// one series per condition
for _, c := range cluster.Status.Conditions {
    v := 0.0
    if c.Status == metav1.ConditionTrue {
        v = 1
    }

    clusterStatus.WithLabelValues(
        uid, name, string(c.Type),
    ).Set(v)
}
```

Declare once, register once, set every loop. **The condition name is a label, not a variable** — so a seventh condition needs no code here at all.

PATTERN 3

0 and 1 give you boolean algebra

Question	PromQL	What it is
All stages passed?	min by (cluster_uid)	AND
Anything passing?	max by (cluster_uid)	OR
Fleet pass rate per stage?	avg by (condition)	mean of 0/1
How many exist?	count(..._info)	cardinality

Don't count your conditions. **min** them.

One condition, six transformations

- 1 YAML type: ArgoAppReady, status: "False"
- 2 Go NewGaugeVec(..., []string{"cluster_uid","condition"})
- 3 Series tenant_cluster_status{cluster_uid="tuv...", condition="ArgoAppReady"} 0
- 4 Scrape OTel collector → VictoriaMetrics or any Prometheus compatible server
- 5 PromQL min by (cluster_uid, name) (tenant_cluster_status)
- 6 Column a Yes / No cell in a table

Demo!



Your operator already knows why it's stuck.

It writes conditions every reconcile. Just export them!

One gauge · condition as a label · min to aggregate

Questions?

LET'S CONNECT!



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